



CONTENT



RESTORATION



AUTHORSHIP



FAMILIARITY



TEST

● ● ●
Name: Roy Batty

Student Number: N6MAA10816

**[01] I've seen things you people wouldn't believe. [02] Attack ships on fire
off the shoulder of Orion. [03] I watched C-beams glitter in the dark near the
Tannhäuser Gate. [04] All those moments will be lost in time, like tears in rain.**

● ● ●

Name: Roy Batty
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Sentence Intruders & Synonym Replacements

Extra sentences have been added. Circle the added sentence numbers.
Five words have been replaced. Find the replacements and provide the originals.
Incorrect sentence numbers (−1 point). Incorrect word choices (0 points).

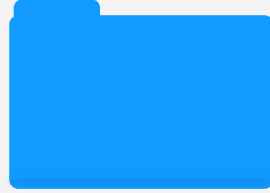
[01] I've seen things you people wouldn't believe. [02] These experiences include witnessing combat ships aflame beyond the reach of conventional rescue operations. [03] Attack ships on fire off the edge of Orion. [04] I observed C-shafts glitter in the dark near the Tannhäuser Gate. [05] All those moments will be lost in time, like weeping in downpour. [06] These fleeting experiences underscore the fragile impermanence of memory amid the relentless passage of time.

part of speech	noun	noun	verb	noun	noun
replacements					
originals	b	t	w	s	r



The Problem

LLM-influenced
Writing may
not reflect
learning
outcomes



Potential Solutions

Artifact-based
approaches &
process-based
approaches



Research Methods & Procedures

within subjects
design;
handwritten text
and unseen LLM
text baselines



Results & Discussion

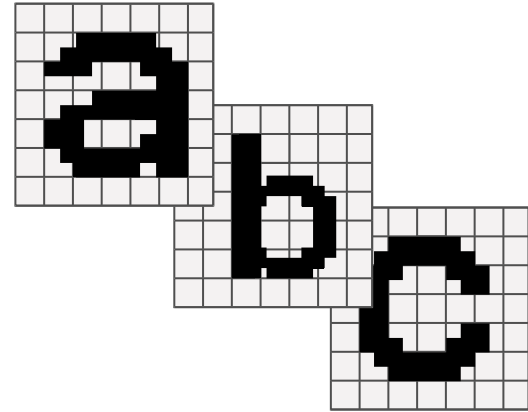
non-overlapping
performance
distribution,
large effect size

The Problem

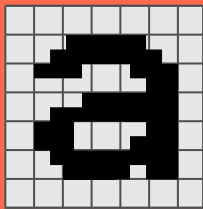
Authorship

When LLMs (e.g., ChatGPT) can generate academic text, writing no longer reliably indicates authorship or learning.

- Students cannot clearly judge their own contribution to LLM-assisted work.
- Institutional responses (bans, detection, disclosure) are unreliable and hard to enforce.

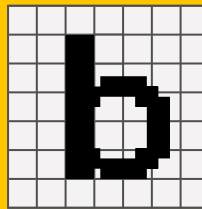


Artifact-based Detection



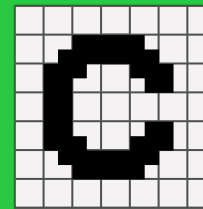
Classifier-based Detection

- + Fast and scalable
- + Trained on human vs. AI text data
- Probabilistic, not proof of authorship
- Sensitive to context (domain, population)



Stylometry-based Detection

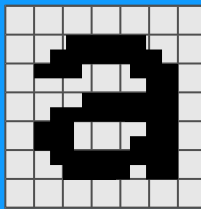
- + Interpretable linguistic features
- + Can capture consistent writing patterns
- Assumes stable individual style
- Weakened by editing or AI assistance



Transformer-based Detection

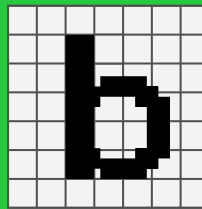
- + Leverage contextual and semantic signals
- + Strong performance in controlled settings
- Dependent on training data/conditions
- Limited generalizability across contexts

Evading Artifact Detection



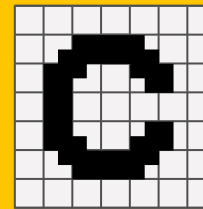
Rewriting attacks

- Paraphrasing disrupts detection
- Accessible paraphrase tools
- Shifts lexical/syntactic patterns



Editing attacks

- Small surface changes undermine model predictions
- May not affect readability
- Low-effort, high-impact evasion



Prompt attacks

- Prompt variation produces alternative outputs
- Iterative refinement reduces detectability

Authorship as a Process



Memory

Authorship is **not inherent** in the text, but reflects a writer's **cognitive relationship** with the text, as shaped by **decisions** made during **planning, drafting, and revision**.

Planning

- Organizing ideas
- Setting goals for expression
- Encoding intended meaning & rhetorical structure

Drafting

- Transforming ideas into text forms
- Actively selecting words & structures
- Encoding formulation decisions

Revision

- Evaluating and refining lexical and structural choices
- Aligning text with intended meaning and goals
- Re-evaluating prior decisions

Checking the Writing Process

Words

- Writers encode specific word choices during planning, drafting, and revision processes
- Writers should notice when their carefully selected words are replaced with synonyms
- CRAFT asks students to identify 5 original and 5 replacement words (10 points)
- Hints provided (first letter, part of speech); no penalty for incorrect responses

Structure

- Writers encode discourse and organizational decisions during planning, drafting, and revision processes
- Writers should notice when sentences do not fit their structure
- CRAFT asks students to identify four (4) LLM-generated intruder sentences (4 points)
- Incorrect selections are penalized (-1 point) to discourage guessing

Research Methods & Procedures

Human Condition

- Participants compose sixteen (16) sentences by hand on paper, devices are prohibited
- Participants transcribe their text and then a Python script creates a personalized CRAFT test for each participant
- About 15 minutes after writing, participants restore their lexical and structural decisions
- Participants have ten (10) minutes to complete the test

LLM Condition

- An LLM generates sixteen (16) sentences on the same topic as the handwritten text
- Generated text is used to create a personalized CRAFT test using the same Python script
- With no knowledge of the text, participants try to identify lexical and structural changes
- Participants have ten (10) minutes to complete the test

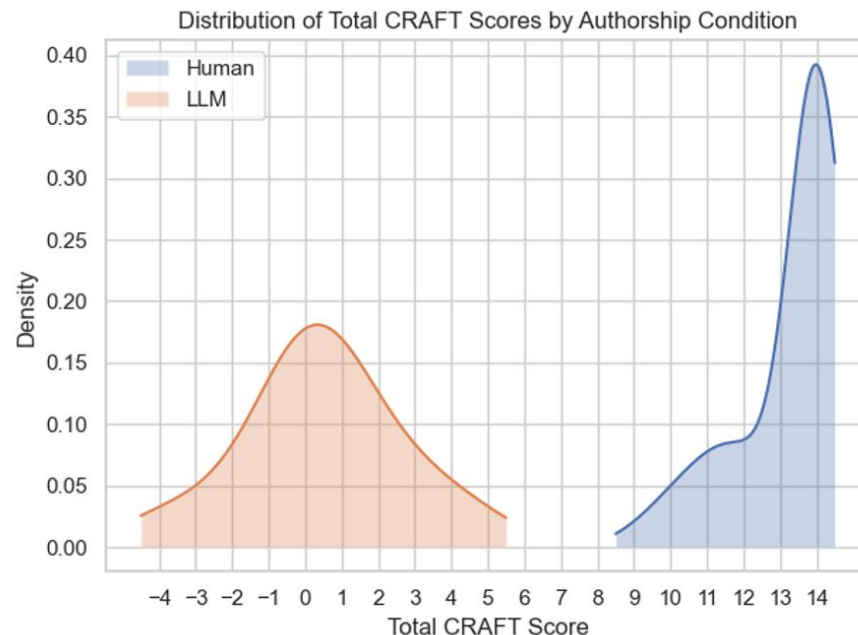
The Results

No overlap

Table 1
CRAFT Performance by Condition

Measure	Human M (SD)	LLM-generated M (SD)	ΔM	Cohen's d	95% CI	N
Structural Restoration	3.97 (0.26)	-0.32 (1.32)	4.28	3.16	[3.93, 4.63]	60
Lexical Restoration	9.28 (1.33)	0.78 (1.30)	8.50	4.65	[8.03, 8.97]	60
Total	13.25 (1.36)	0.47 (2.09)	12.78	5.34	[12.16, 13.40]	60

Note. Values are means with standard deviations in parentheses. ΔM = mean paired difference (Human - LLM-generated). Cohen's d reflects paired effect size. CI = confidence interval.



Discussion & Future Directions

Writing increasingly blends human & machine contributions; research should **map this space** using process-based measures

- Clear separation between human and LLM baselines may support measurement of future work, including hybrid texts
- Lexical and structural measures capture complementary aspects of writing, probing processes rather than surface features
- Limitations: effort effects and generalizability
- Future work: binary detection → graded authorship (incl. hybrid text)
- Practical use: baseline profiles support evidence-based, non-adversarial assessment, and writing process dialogue

Authorship is cognitive not textual, as writing encodes retrievable lexical and structural decisions that can be directly assessed.

Theoretical Contribution

Questions? Please ask!



Citation: Rose, R. (2026). Process-Based Authorship Assessment Across Human and LLM-Generated Writing: The Content Restoration Authorship Familiarity Test (CRAFT). Manuscript in press.

Preprint: <http://ssrn.com/abstract=6879243>



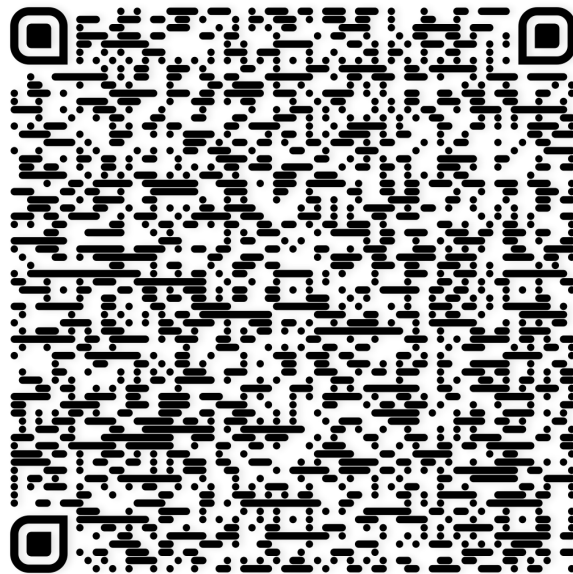
Email

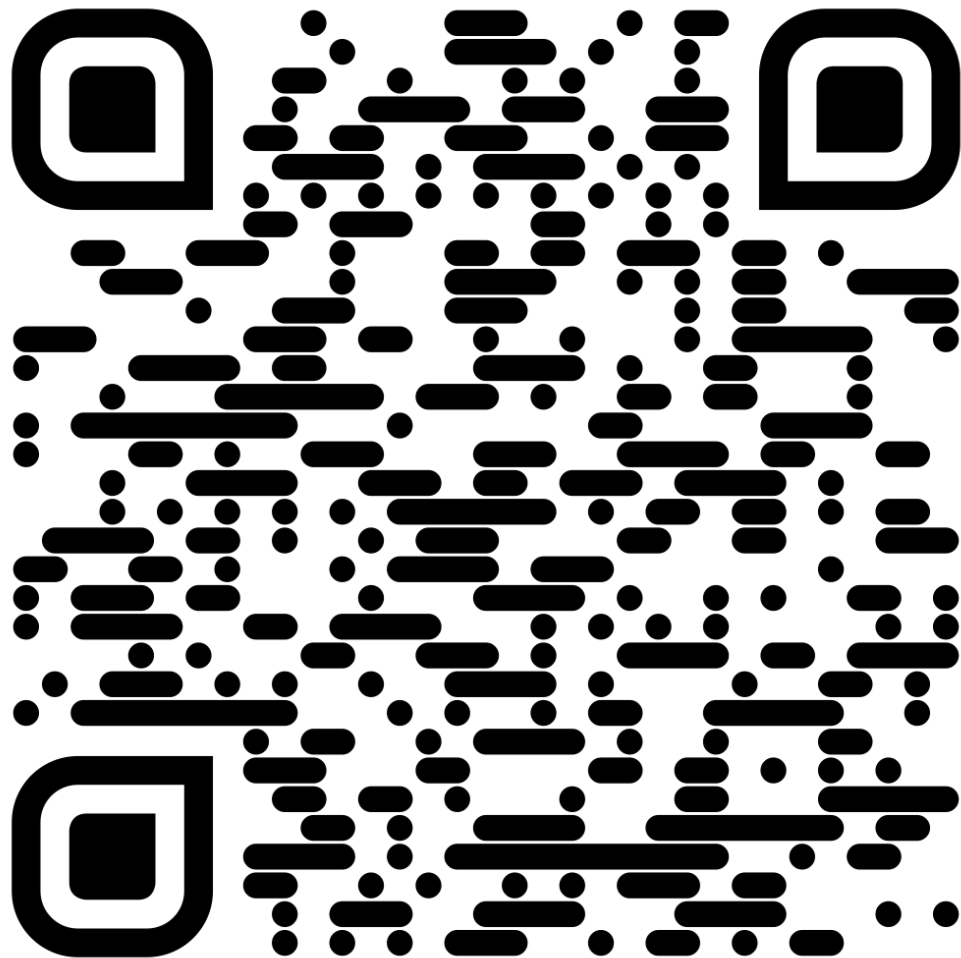
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Github





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